Supporting Information: Supporting Information for "Resistance-Aware Computational Prioritization of Antimalarial Leads from African Natural Products: Docking, Chemical-Space Descriptors, and Exploratory Molecular Dynamics"

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MD scope. The parent-study trajectories comprise a targeted, non-membrane explicit-solvent structural follow-up of four wild-type complexes. A separate 16-system Set-C pilot evaluated

Table S1: *Docking enrichment and re-docking accuracy. Values are from corrected parent-study benchmarks using V2 grids. MMV Malaria Box values are consensus-score ROC-AUCs for PfDHFR, PfCRT, and PfATP4; DEKOIS 2.0 used AutoDock Vina for PfDHFR. BEDROC values use* $\alpha = 20$. *Re-docking RMSD is the heavy-atom RMSD between the top-scoring pose and the co-crystallized ligand; the MTX RMSD was approximately* 30 Å.

Target	Method	ROC-AUC	EF1 %	EF5 %	BEDROC	RMSD (Å)
PfDHFR (7F3Y)	Vina redock	–	–	–	–	N/A
	MMV consensus (Vina+DiffDock)	0.924	–	2.57	1.309	–
	DEKOIS (Vina only)	0.450	–	0.00	–	–
PfCRT (6UKJ)	MMV consensus (Vina+DiffDock)	0.971	–	1.11	9.434	–
PfATP4 (9N10)	MMV consensus (Vina+DiffDock)	1.000	–	2.00	1.810	–
PfClpR (4GM2; not PfClpP)	MMV enrichment	not evaluated	–	–	–	–

Target identity. PDB 4GM2 is PfClpR rather than PfClpP; it is not used as PfClpP validation.

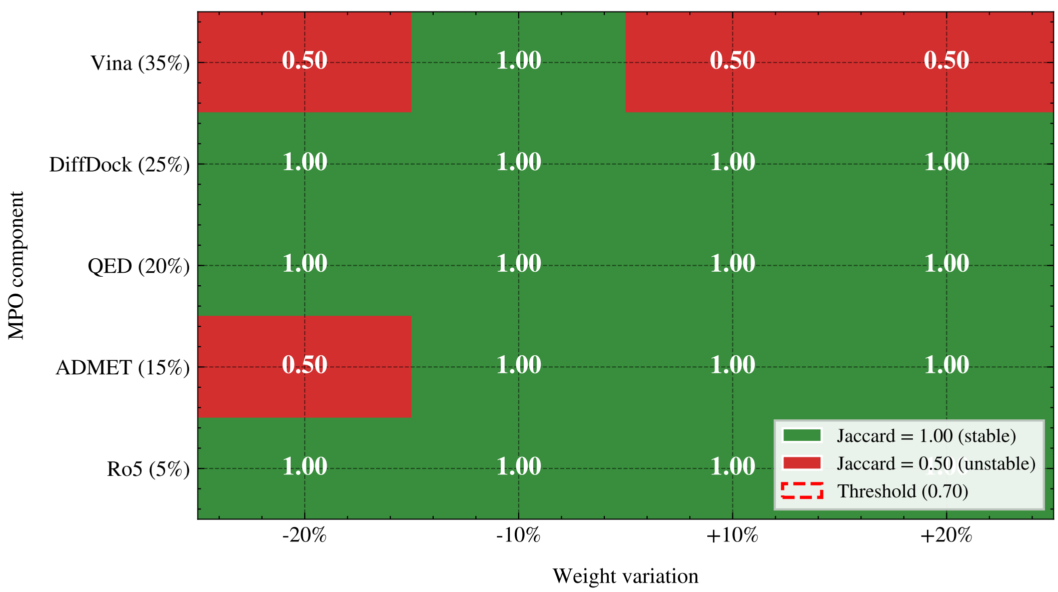

Figure S1: *MPO sensitivity analysis. Each cell shows the Jaccard similarity of the top-20 list after varying one MPO weight by* ±20 %. *Green* (1.00) *indicates no change; red* (0.50) *indicates substantial reordering.*

Table S2: *MM-GBSA results for the four parent-study complexes. These systems are distinct from the 17 Set-C candidates. The interpretable PfCRT–214 result is reported as mean ± standard error of the mean (SEM) from* 101 *snapshots; dissociated or conversion-invalid systems are listed as N/A. Mutant RRS was calculated only for Set-C docking systems.*

Ligand	Target	ΔG_{bind}	SEM	Interpretation
164	PfClpR-labelled cohort*-WT	–	–	N/A; ligand dissociated‡
201	PfDHFR-WT	–	–	N/A; ligand dissociated‡
214	PfCRT-WT	−18.25	0.40	Interpretable (gmx_MMPBSA [S1])
438	PfATP4-WT	–	–	N/A; conversion corrupted§

*PDB 4GM2 is PfClpR rather than PfClpP; the 164-labelled system is therefore not structural validation of PfClpP.
‡Ligand dissociated during production; no binding free energy is inferred.
§Excluded due to two-chain topology incompatibility with CHARMM36 [S2]-to-AMBER conversion.

Table S3: *Predicted ADMET profile of the* 17 *set-C polypharmacology candidates (ADMET-AI, parent-library screening). All values are machine-learning predictions; experimental ADMET profiling and three-tool concordance were outside the scope of this study. Across candidates, LogS ranged from* 0.34 *to* 0.73*; CYP3A4 inhibition was* ≥ 0.5 *for* 7/17 *candidates, Caco-2 permeability was* ≥ 0.5 *for* 15/17*, and Clint*$_h$ *ranged from* 0.48 *to* 0.63.

Compound	ADMET-AI LogS	CYP3A4 prob.	Caco-2 prob.	Clint$_h$
PP-01	0.34	0.39	0.63	0.60
PP-02	0.46	0.97	0.86	0.55
PP-03	0.43	0.62	0.95	0.55
PP-04	0.40	0.58	0.92	0.63
PP-05	0.61	1.00	0.35	0.51
PP-06	0.54	0.90	0.97	0.60
PP-07	0.36	0.02	0.75	0.62
PP-08	0.59	1.00	0.41	0.56
PP-09	0.65	0.13	0.81	0.57
PP-10	0.49	0.13	0.93	0.62
PP-11	0.39	0.06	0.96	0.58
PP-12	0.49	0.00	0.98	0.48
PP-13	0.57	0.97	0.98	0.53
PP-14	0.66	0.05	0.68	0.49
PP-15	0.66	0.49	0.95	0.51
PP-16	0.54	0.25	0.91	0.50
PP-17	0.73	0.01	0.82	0.50

Table S4: *Candidate-cohort ACSI weight sensitivity. One ACSI component weight was varied by* $\pm 20\,\%$ *and the remaining weights were rescaled proportionally to preserve a unit sum. Mean and standard deviation refer to the resulting ACSI values across the* 17 *candidates. Rank stability is measured by Spearman correlation over all candidates and Jaccard overlap of the top five candidates with the baseline ranking. Because the full-library component matrix was unavailable, this analysis applies only to the 17-candidate Set-C cohort; it does not assess the full-library top-20 ranking.*

Weight varied	Variation (%)	New weight	Mean ACSI	SD	Spearman ρ	Top-5 Jaccard
D_{DrugBank}	-20	0.320	0.512	0.149	0.977 9	0.666 7
D_{DrugBank}	+20	0.480	0.574	0.159	0.933 8	0.666 7
D_{ANPDB}	-20	0.200	0.568	0.149	0.916 7	0.666 7
D_{ANPDB}	+20	0.300	0.518	0.158	0.980 4	0.666 7
f_{sp^3}	-20	0.160	0.552	0.152	0.951 0	0.666 7
f_{sp^3}	+20	0.240	0.534	0.153	0.958 3	0.666 7
NPL	-20	0.120	0.535	0.160	0.963 2	0.666 7
NPL	+20	0.180	0.551	0.145	0.980 4	1.000 0

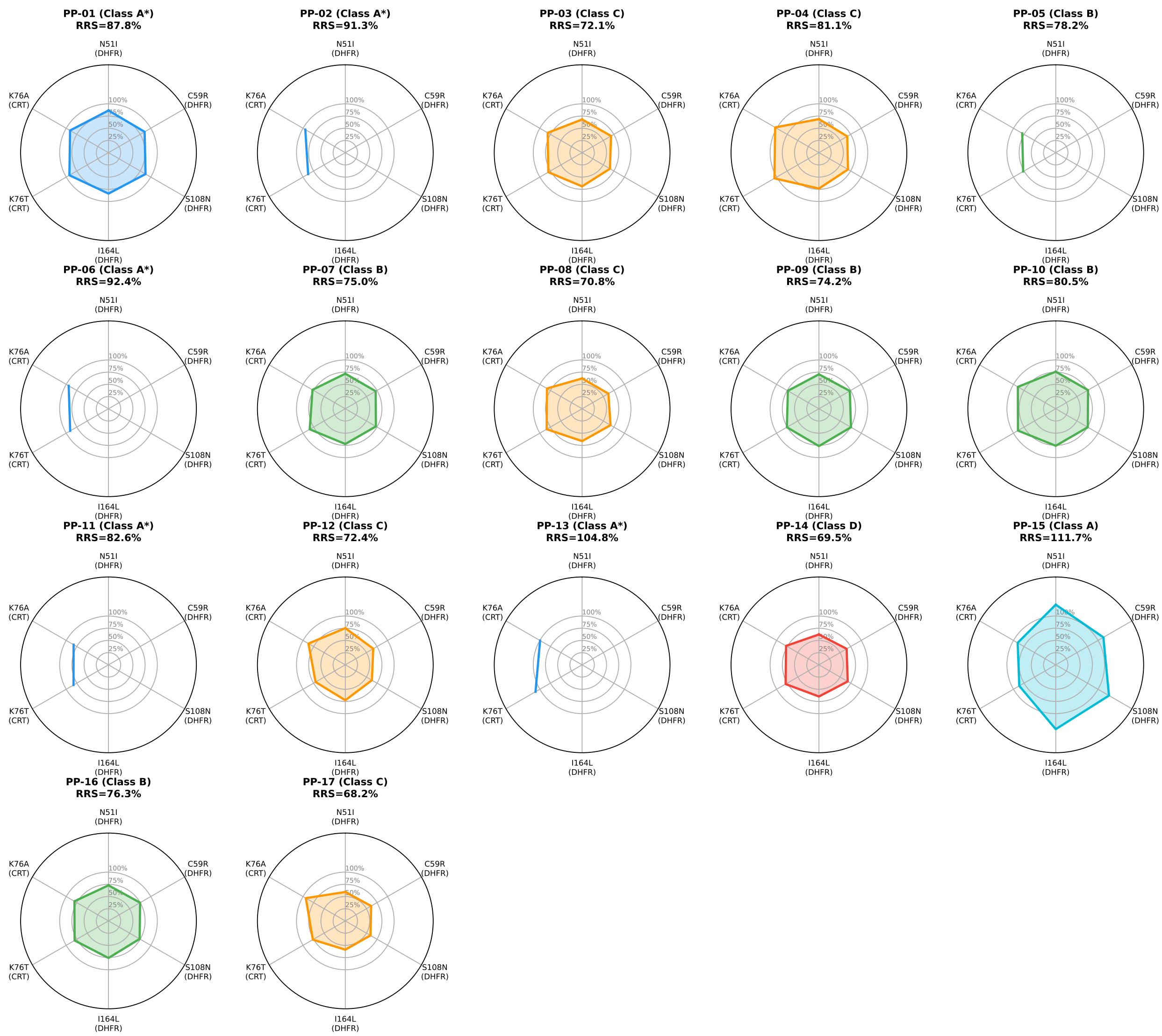

Figure S2: *Per-compound, docking-derived RRS profiles across six resistance mutations for the 17 Set-C candidates. Missing axes denote an ineligible target under the* $5.0\,\mathrm{kcal\,mol^{-1}}$ *wild-type docking-score threshold and are not treated as zero. Concentric rings indicate RRS values of* $50\,\%$, $100\,\%$, *and* $150\,\%$*; the plot does not encode absolute binding potency.*

PP-01 and PP-02 against the six-mutant panel for 10 ns per system. Its MD-RRS and MM-GBSA estimates are summarized in Table S18 and compared with docking RRS in Figure S3. Neither design estimates long-timescale residence or replicate-level uncertainty; the evidence boundary is summarized in Tables S14 and S15.

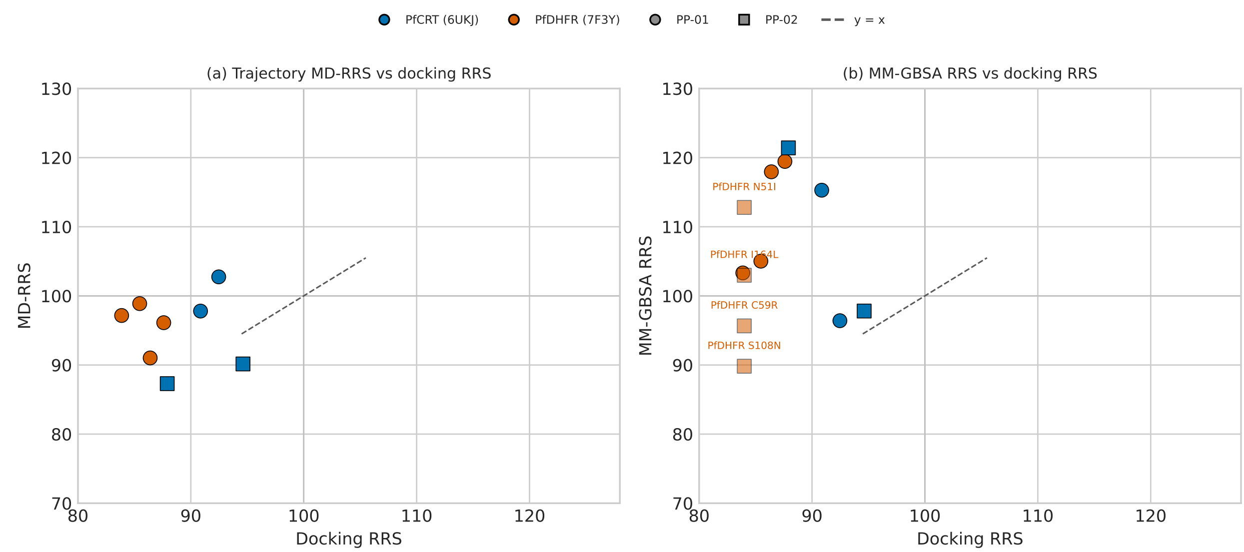

Figure S3: *Trajectory-derived and docking-derived RRS values for the Set-C pilot, normalized to the corresponding WT value of* 100*. Panel (a) compares mean minimum-distance ratios with docking RRS for eight mutant systems; panel (b) compares MM-GBSA RRS with docking RRS. Four PP-02 PfDHFR systems lack a docking WT reference. Docking predicted lower mutant scores in the comparable systems, whereas the trajectory-derived estimates did not reproduce that pattern. Neither estimate measures biological resistance or thermodynamic affinity.*

Table S5: *Stable Plasmodium falciparum target identifiers and mutation contexts used in this study. The identifiers define the targets and mutation states; they do not provide pathway-level or experimental evidence.*

Target	ID	Context	Mutation/evidence boundary
PfDHFR	`PF3D7_0417200`	7F3Y WT template	N51I/C59R/S108N/I164L; antifolate states
PfCRT	`PF3D7_0709000`	6UKJ WT template	K76T/K76A resistance states
PfATP4	`PF3D7_1211900`	9N10 parent-study structure	Annotation only; outside Set-C panel
PfClpP	`PF3D7_0307400`	Proteolytic Clp subunit	Not assigned to 4GM2; no PfClpP claim
PfClpR	`PF3D7_1436800`	4GM2, PfClpR	4GM2 is PfClpR, not PfClpP

Stable identifiers were retrieved from PlasmoDB/VEuPathDB on 12 August 2026[S5]. The database release/build was not available. The existing PfCRT pH 5.2 redocking sensitivity (100/100 library ligands; $\rho = 0.270$, $p = 0.006551$) was not merged into the Set-C docking-

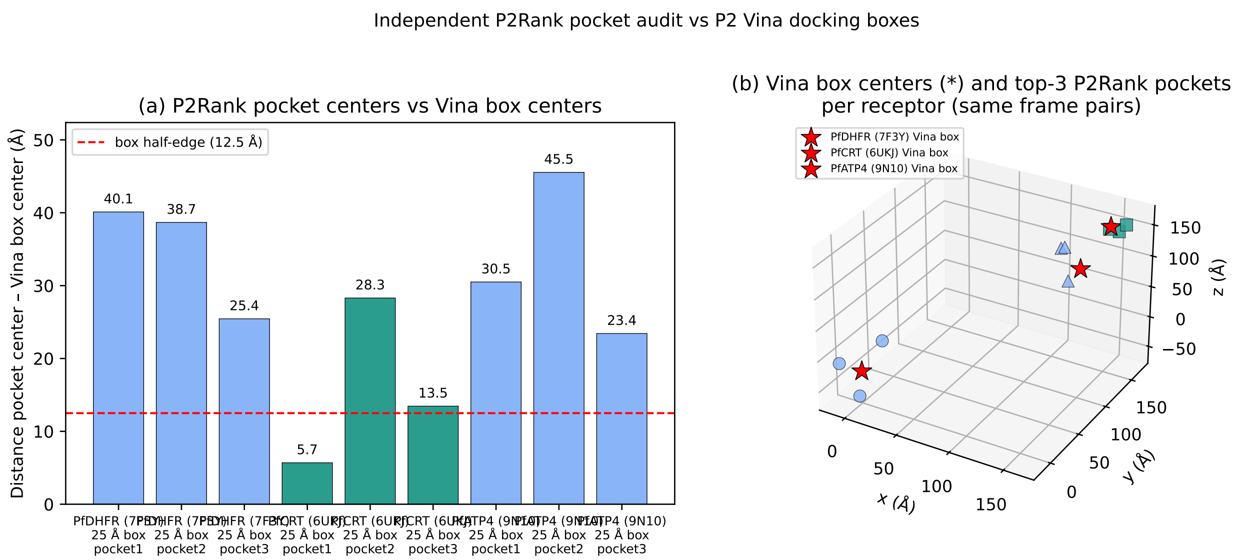

Figure S4: *Independent P2Rank*[S3] *pocket prediction versus the fixed Vina*[S4] *grid boxes used for docking. Panel (a) reports the distance between the top-three P2Rank pocket centers and the Vina box center for the three receptors with a P2 grid (PfDHFR 7F3Y, PfCRT 6UKJ, PfATP4 9N10); the dashed line marks the* 12.5 Å *box half-edge. Panel (b) shows the same centers in 3D (star: Vina box; markers: top-three P2Rank pockets, same frame pairs). For PfCRT (6UKJ), the top P2Rank pocket (score* 162.7*, probability* 0.999*, by far the most ligandable site) lies* 5.7 Å *from the Vina box center, corroborating the docking box. The larger separations for PfDHFR and PfATP4 carry a receptor-frame caveat (the P1 grid configuration may derive from a different preparation than the MD receptor used here) and are recorded as provenance flags rather than docking-quality verdicts. P2Rank 2.5.1; exploratory boundary audit only — no canonical score or RRS value is modified.*

RRS cohort. No GO or pathway-enrichment test was performed because an independent multi-gene targetome and prespecified parasite-proteome background were unavailable[S6].

Table S6: *Questions, estimands, and interpretation boundaries. The Set-C docking cohort contains 17 candidates and 136 WT/mutant systems selected after prespecified prioritization filters; 12 candidates have complete eligible PfDHFR/PfCRT panels and define the target-balanced primary RRS analysis, whereas 5 are PfCRT-only sensitivity records. The parent-study MD cohort is non-overlapping. The Set-C MD pilot met the predefined trajectory-quality criteria for all 16 systems and is analyzed separately; its MD-RRS and MM-GBSA estimates are not merged into the docking-RRS classification.*

Evidence layer	Cohort	Systems	Primary analyses	Interpretation boundary
Set-C docking	PP-01–PP-17	136 (12 complete; 5 PfCRT-only)	WT/mutant docking; target-specific RRS, ACSI, PNS	Filtered prioritisation cohort; not measured binding or MD-RRS
Parent-study MD	Ligands 164, 201, 214, 438	4 WT complexes × 10 ns	RMSD, contacts, bound-state assessment, MM-GBSA where valid	Structural check; one interpretable MM-GBSA estimate (PfCRT–214)
Set-C MD pilot	PP-01/PP-02	16 systems (all met trajectory criteria)	Trajectory geometry, MD-RRS, MM-GBSA	Single-replicate structural follow-up; separate from docking-RRS classification
Full Set-C MD-RRS	PP-01–PP-17	136 planned systems	Full trajectory QC and MD-RRS	Not available because the complete cohort was not simulated

Table S7: *Sensitivity of the PNS ranking to the imputed PfCRT centrality. The reference value is the observed network-mean imputation ($C_{\mathrm{PfCRT}} = 0.151$). Alternative values were zero, one-half, one-and-a-half, and twice the reference value. Spearman ρ is calculated against the reference 17-candidate ranking; all candidates have two target entries in the PNS record. The ranking is sensitive to this network-data choice; target essentiality and biological polypharmacology were not assessed.*

Imputation	C_{PfCRT}	Spearman ρ	Minimum PNS	Maximum PNS	n
Zero	0.0000	0.9716	0.300	5.147	17
One-half canonical	0.0755	0.9975	0.674	5.573	17
Canonical network mean	0.1510	1.0000	1.047	6.000	17
One-and-a-half canonical	0.2265	0.9755	1.421	6.426	17
Twice canonical	0.3020	0.9632	1.795	6.853	17

Secondary calibration and ranking analyses

Tartarus docking calibration

The Tartarus analysis used QuickVina[S7], an AutoDock Vina[S4] derivative, on 19 913 primary leads against PDB entries 1SYH, 6Y2F, and 4LDE. The composite MPO comparison included up to 17 211 complete observations. The composite association was $\rho = 0.0129$

($p = 0.091\,3$); target-specific associations were small and do not imply score equivalence or independence.

Table S8: *Tartarus docking calibration: Spearman correlations between QuickVina scores and weighted MPO scores.*

Score	Spearman ρ	p-value	n	Interpretation
Composite (mean of 3 targets)	0.012 9	9.13×10^{-2}	17 211	No meaningful association
1SYH (PfDHFR)	0.065 5	8.18×10^{-18}	17 205	Small effect
6Y2F (PfCRT)	−0.004 0	6.04×10^{-1}	17 211	Null
4LDE (A2A receptor)	−0.141 5	1.17×10^{-77}	17 211	Small effect

Tartarus cross-docking correlation

Only 24 molecules overlapped between the Tartarus and project docking records. Because the structures were not matched, this comparison cannot calibrate the docking scores experimentally.

Table S9: *Tartarus independent-tool comparison for the 24-molecule overlap.*

Tartarus	Project metric	n	ρ	p	Interpretation
Composite	4GM2/PfClpR	24	0.246	2.47×10^{-1}	No
Composite	6UKJ/PfCRT	24	0.075	7.28×10^{-1}	No
Composite	7F3Y/PfDHFR	24	−0.110	6.10×10^{-1}	No
Composite	9N10/PfATP4	24	0.118	5.83×10^{-1}	No
1SYH/DHFR	7F3Y/PfDHFR	24	−0.442	3.04×10^{-2}	Exploratory

Complete ACSI components

ACSI was calculated as

$$\mathrm{ACSI} = 0.40 D_{\mathrm{DrugBank}} + 0.25 D_{\mathrm{ANPDB}} + 0.20 f_{sp^3} + 0.15\mathrm{NPL}.$$

Components were min–max normalized over the 17-candidate cohort. Two candidates exceeded ACSI > 0.70, and the cohort mean was 0.543. Varying one weight at a time by

±20 % gave Spearman correlations of 0.916 7–0.980 4 and top-five Jaccard overlaps of 0.666 7–1.000 0.

PNS ranking and imputation

The STRING[S8] network used confidence threshold 700; PfCRT centrality was imputed with the network mean, 0.151. The full ranking is listed in the table. Replacing the imputed value with zero, one-half, one-and-a-half, or twice the reference value gave rank correlations of 0.963 2–1.000 0. This result reflects sensitivity to a network-data choice, not biological target importance.

Table S10: *PNS ranking and RRS class. PfCRT-only denotes candidates without an eligible PfDHFR WT denominator for primary RRS.*

Candidate	PNS	ACSI	RRS class
Rank 1	6.00	0.599	C
Rank 2	4.71	0.308	C
Rank 3	4.53	0.567	C
Rank 4	4.48	0.603	B
Rank 5	4.48	0.551	B
Rank 6	4.45	0.173	B
Rank 7	4.45	0.459	A*
Rank 8	4.36	0.743	D
Rank 9	4.36	0.605	C
Rank 10	4.34	0.594	C
Rank 11	4.20	0.601	B
Rank 12	3.50	0.576	A
Rank 13	2.32	0.597	PfCRT-only
Rank 14	1.23	0.823	PfCRT-only
Rank 15	1.14	0.387	PfCRT-only
Rank 16	1.10	0.458	PfCRT-only
Rank 17	1.04	0.589	PfCRT-only

Predicted ADMET

DiffDock–Vina and activity-model comparisons

In the 65 856-molecule library, Vina[S4] scores and DiffDock[S9] confidence were only weakly related overall. A Benjamini–Hochberg analysis of 18 pairwise activity comparisons across compound groups using three Ersilia[S10] models found no adjusted-significant differences; all rank-biserial effect sizes were negligible. These analyses provide no evidence of biological activity.

The ADMET-AI[S11] endpoint table for all 17 candidates is given in Table S3. These predictions were not experimentally validated or compared across tools.

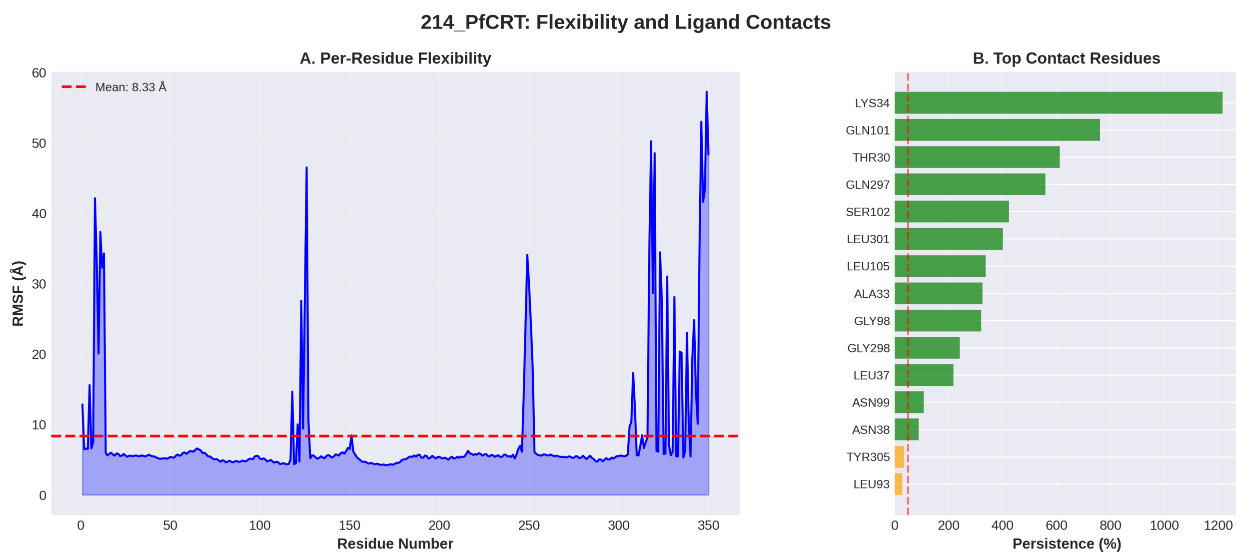

Figure S5: *Per-residue flexibility and binding-contact persistence for the PfCRT–Ligand 214 complex. LYS34 is the main contact anchor; contact persistence is an aggregate count, not a fractional occupancy.*

State-specific summaries. For PfDHFR, mean RRS values were 74.7 (N51I), 73.7 (C59R), 75.3 (S108N), and 76.8 (I164L). For PfCRT, the corresponding means were 85.4 (K76T) and 87.0 (K76A). These target-specific summaries describe score retention and do not measure biochemical resistance.

RRS threshold sensitivity. Across the prespecified threshold grid, the baseline available-target class counts were unchanged when the WT eligibility threshold was varied from

Table S11: *Endpoint replicate consistency for the PP-01 PfCRT wild-type pillar. Two independent* 10 ns *productions executed under the identical MDP and analysis chain (GB^{OBC2} igb=5,* 0.15 M *salt,* 100 *snapshots) are compared. Tabulated spreads are the within-trajectory SD and the SEM over stored snapshots; the manuscript-wide quoted uncertainty elsewhere is the SEM. The SEM is informational here because MD snapshots are temporally autocorrelated. The between-replicate offset (*$|\Delta| = 2.01\,\mathrm{kcal\,mol^{-1}}$*) lies within one within-trajectory standard deviation of either run (inter-replicate mean* $(-29.61 \pm 1.42)\,\mathrm{kcal\,mol^{-1}}$*). Because the snapshots are temporally autocorrelated, the SEM does not quantify replicate-level uncertainty. The offset is therefore used as a conservative empirical noise floor of* $2.0\,\mathrm{kcal\,mol^{-1}}$ *for interpreting single-replicate mutant–wild-type endpoint contrasts.*

Replicate	Production	ΔG_{bind} ($\mathrm{kcal\,mol^{-1}}$)	SD	SEM
R1	production run 1	−30.61	4.12	0.41
R2	independent production run	−28.60	4.35	0.43

Table S12: *Sensitivity of the available-target RRS classes to the WT eligibility and retention thresholds. Counts are reconstructed from the versioned threshold-sensitivity output for all 17 candidates. The prespecified thresholds used in the main analysis are shown in bold. These are classification-stability diagnostics; they do not calibrate the thresholds against experimental resistance measurements.*

WT threshold	Lower retention	Upper retention	A*	A	B	C	D
5.0	70	80	**5**	**1**	**5**	**5**	**1**
5.0	75	85	3	1	4	3	6
5.0	80	90	2	0	4	2	9
4.0	70	80	5	1	5	5	1
6.0	70	80	6	0	5	5	1
7.0	70	80	6	0	5	5	1
7.0	75	85	4	0	4	3	6
7.0	80	90	2	0	4	2	9

Table S13: *Target-specific docking-score retention in the eligible candidates. Values are summarized across candidates with an eligible WT denominator for the indicated target. The target-specific class counts are descriptive and are not equivalent to the primary two-target classification.*

Target	Eligible candidates	Mean RRS	Median RRS	Range	Class A*/A	Class B/C/D
PfDHFR	12	75.1	70.8	59.7–123.4	2	10
PfCRT	17	86.2	83.5	74.7–104.8	10	7

Table S14: *Evidence scope and next validation. The computational results support prioritization and structural hypotheses within the stated cohorts; they do not substitute for biochemical, biophysical, phenotypic, or clinical measurements.*

Evidence layer	Supported interpretation	Next validation needed
Docking-RRS	Relative retention of empirical docking scores across the selected mutant panel; computational triage only	Matched biochemical activity measurements for wild-type and mutant PfDHFR/PfCRT
ACSI/PNS	Cohort-relative chemical-space and network-weighted ranking dimensions	Larger, independently selected panels and experimentally anchored target/network data
Parent-study MD	Ligand retention under the declared geometric criterion and system-specific endpoint assessment	Replicated, membrane-aware simulations and orthogonal binding or transport assays
Set-C MD pilot	Secondary trajectory-derived comparison for PP-01 and PP-02; single replicate per system	Independent mutant/wild-type replicates with prespecified convergence and uncertainty analyses
Biological polypharmacology	Not measured by the present computational estimands	Target engagement plus parasite-phenotype and mechanism-of-action experiments

Table S15: *Scope of the targeted resistance panel. The panel was designed to test resistance-aware ranking across selected molecular contexts; it was not designed to estimate resistance prevalence, reconstruct the parasite targetome, or establish pathway-level polypharmacology.*

Dimension	Covered in this study	Not covered
PfDHFR	WT plus N51I, C59R, S108N, and I164L docking states	Other resistance haplotypes, prevalence, and experimental activity
PfCRT	WT plus K76T and K76A docking states; neutral-pH protocol	Full haplotype diversity, acidic-vacuole conditions, and transport measurements
PfATP4	One parent-study structural context	Mutant-state panel and resistance inference
PfClpR	One parent-study context associated with PDB 4GM2	PfClpP validation and broader proteostasis target coverage
Chemical cohort	17 filtered Set-C candidates; 12 define the complete two-target primary analysis	Representative sampling of the 65,856-molecule library
Biological outcome	Computational docking, trajectory, and endpoint summaries	Direct target engagement, parasite phenotype, and pathway-level mechanism

Table S16: *Lightweight robustness and transfer audits for the selected Set-C cohort. These post-selection diagnostics assess sensitivity of the computational ranking and class definitions; they do not establish biochemical affinity, biological resistance, target engagement, or mechanism of action.*

Audit	Result	Interpretation boundary
Leave-one-candidate-out	Complete two-target $n = 12$; PNS–RRS ρ ranged from -0.3182 to 0.0273; ACSI–RRS ρ ranged from -0.7636 to -0.3182	Rank associations are sensitive to candidate removal and remain exploratory
Target-stratified correlations	PfDHFR: PNS–RRS $\rho = -0.3497$; PfCRT: $\rho = -0.0417$	Target-specific patterns are not interchangeable with the pooled estimand
Score perturbation	$\pm 1\,\mathrm{kcal\,mol^{-1}}$ perturbations generated 272 candidate–score records; class unchanged in 212/272 (77.9 %)	Operational RRS classes show threshold sensitivity, especially to WT-score perturbation
Threshold sensitivity	A prespecified threshold grid was retained unchanged in the source record	Thresholds are triage rules, not experimentally calibrated categories
Related ChEMBL transfer	The public ChEMBL panel used by the companion study was verified against its source record; no P2-specific external panel was available	Related-project transfer evidence is not an independent P2 replication or experimental validation
External docking replication	39 ligands, 8 PfDHFR/PfCRT states, 312/312 finite records after declared EXT-039 embedding exclusion; bootstrap mean RRS 100.45 (99.59–101.32), Class-A fraction 0.974 (0.923–1.000)	Computational docking sensitivity analysis only; not experimental validation or an MD estimate

Table S17: *Use of companion P1 evidence in P2. P1 results are included only as methodological or chemical-space context. They do not validate P2 mutant-state RRS, biological resistance, or polypharmacology.*

P1 evidence	Permitted use in P2	Boundary
Redocking and enrichment benchmarks	Context for pose reproduction and general ligand-ranking behavior of the docking workflow	Does not validate mutant-state score retention or RRS
Physicochemical and drug-likeness profiles	Context for describing the upstream chemical library and the selected PP-01–PP-17 cohort; the cohort audit found 17/17 exact SMILES matches	Shared candidate identity and provenance preclude independent-replication claims; properties are not biological activity measurements
Chemical-space and scaffold analyses	Context for the origin and diversity of the parent library	Set-C remains a filtered prioritization cohort, not a representative sample
MPO sensitivity	Context for the upstream selection procedure and its local stability	Does not test RRS or network-weighted ranking
P1–P2 shared candidates or provenance	Used only with explicit cohort and provenance labeling	Not an independent replication when candidates or source workflows overlap

$4.0\,\mathrm{kcal\,mol^{-1}}$ to $5.0\,\mathrm{kcal\,mol^{-1}}$. Tightening the retention thresholds from 70 %/80 % to 80 %/90 % increased the number of Class D candidates from 1 to 9. This sensitivity reflects the operational nature of the docking-based classes and does not calibrate them against experimental resistance measurements.

Table S18: *MM-GBSA endpoint estimates for the Set-C pilot.* ΔG_{bind} *is the mean over* 100 *snapshots; the reported* $\mathrm{SD}_{\mathrm{prop}}$ *is the propagated within-trajectory standard deviation, not a SEM or replicate uncertainty. MMG-RRS is the ratio of absolute mutant and WT endpoint magnitudes, with WT = 100 %. It is not a free-energy resilience measure. n/a denotes the absence of a docking WT reference for PP-02 PfDHFR. Two systems required trajectory reconstruction before endpoint analysis because of periodic-boundary discontinuities. Values above* 100 % *indicate retention within endpoint uncertainty, not increased affinity; eight of the twelve mutant systems exceed this value.*

Candidate	Target	State	ΔG_{bind} ($\mathrm{kcal\,mol^{-1}}$)	SD_prop	MMG-RRS	MD-RRS	Docking RRS
PP-01	PfCRT	K76A	−35.29	1.17	115.3	97.8	90.86
PP-01	PfCRT	K76T	−29.51	2.65	96.4	102.8	92.47
PP-01	PfCRT	WT	−30.61	1.65	100.0	100.0	–
PP-01	PfDHFR	C59R	−26.56	1.91	105.0	98.9	85.47
PP-01	PfDHFR	I164L	−26.13	2.34	103.3	97.2	83.87
PP-01	PfDHFR	N51I	−29.83	1.50	118.0	91.0	86.40
PP-01	PfDHFR	S108N	−30.21	3.62	119.5	96.1	87.60
PP-01	PfDHFR	WT	−25.29	2.94	100.0	100.0	–
PP-02	PfCRT	K76A*	−24.53	1.27	97.8	90.2	94.62
PP-02	PfCRT	K76T	−30.45	1.38	121.4	87.3	87.91
PP-02	PfCRT	WT	−25.08	2.56	100.0	100.0	–
PP-02	PfDHFR	C59R	−28.85	1.88	95.7	102.0	n/a
PP-02	PfDHFR	I164L	−31.06	1.52	103.0	97.4	n/a
PP-02	PfDHFR	N51I	−34.02	2.68	112.8	72.2	n/a
PP-02	PfDHFR	S108N	−27.09	1.55	89.8	97.5	n/a
PP-02	PfDHFR	WT*	−30.16	2.06	100.0	100.0	–

Use of Artificial Intelligence

During the preparation of this work the authors used AI assistants to support code development and data analysis scripting. The authors reviewed and edited all AI-assisted output and take full responsibility for the content of this article. No generative AI tool is listed as an author. All computational results, statistical analyses, and quantitative claims were generated and verified by the authors.

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